3W Well Data Viewer

A desktop GUI to inspect the real instances of the 3W dataset

Thiago Tomás de Paula

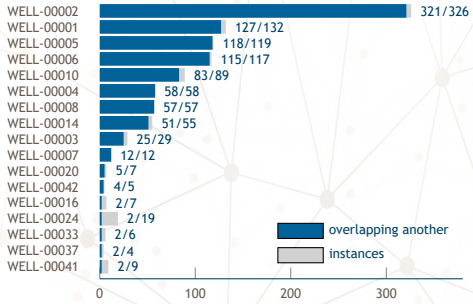
University of Brasília

5th Workshop on the 3W dataset
November 2026

Overlapping samples

Timelines page ► How are the instances of one well distributed in time?

17(42.5%) wells have overlaps, totalling 986(88.11%) instances



Example: instances overlapping in WELL-00004.



Worries:

- Overlapping samples may land in **train and test**, causing leakage.
- Duplicated stretches pull **clusters** tighter, inflating scores.

Thankfully, 3W 2.0.0 has no label collisions, making joins principled and avoiding leakage.

Clustering must still be carefully calculated!

What a joined history is labeled as

Timelines page ► How are the instances of one well distributed in time?

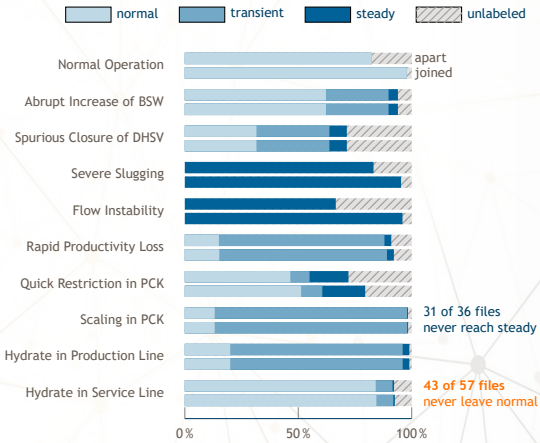
12.3 % ▷ 2.9 %
of unlabels after join

After the join, most unlabeled stretches appear trivial and might be recoverable.

Unlabeled stretch position	*n*	Duration	Share
head of a recording	208	1 h each	86.5 %
seam between equal labels	882	median 76 s, max 56 min	11.5 %
between two labels	16	median 6 min, max 1 h	2.0 %

▸ Appendix

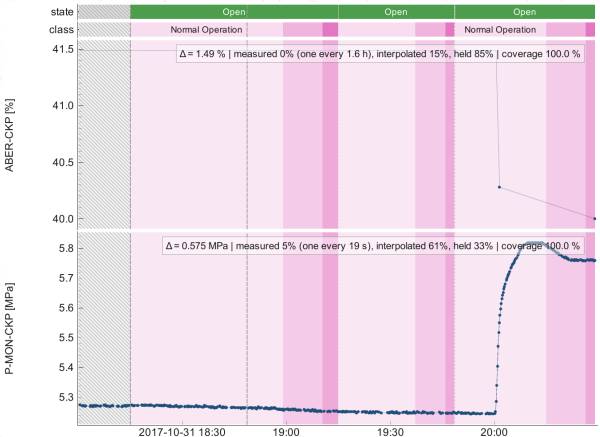
Each fault folder's recorded time, by label, its files apart and joined



Histories whose labels do not add up

Timelines page ► How are the instances of one well distributed in time?

WELL-00004's joined Quick Restriction in PCK instances.



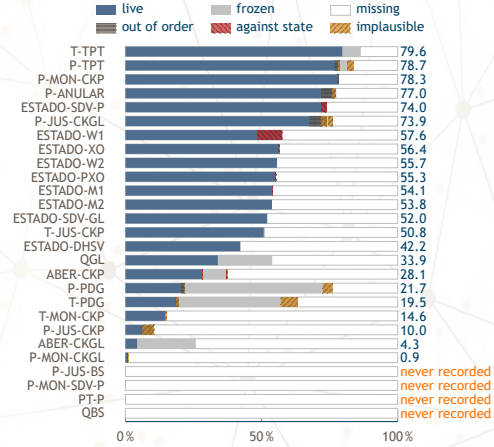
- Joined histories go back to normal **13 times** in 7 wells, **every time at a file boundary**.
- **WELL-00004**, left: three quick restrictions in 75 min, each back to normal 5 to 15 s after its steady state.
- **WELL-00006**: Abrupt Increase of BSW, steady, then normal 30 min later in a Scaling in PCK file.
- The others: Flow Instability (WELL-00001, 00004), Rapid Productivity Loss (00016, 00020), Scaling in PCK (00024), Hydrate in Service Line (00042).

Appendix

What each variable recorded

Availability page ► How well do sensors and states cover the dataset?

See the detailed spread in the ▸ Appendix



PL check P-PDG → P-TPT → P-MON-CKP → *PCK open* → P-JUS-CKP

SL check P-MON-CKGL → *CKGL open* → P-JUS-CKGL ← *M2, W2 open* ← P-ANULAR

Valve checks

Open	M1, W1, SDV-P, PCK open; PXO, XO closed
Restart	M1, W1, SDV-P, PCK open
Shut-In, Closed With Diesel or Gas	one of M1, W1, SDV-P, PCK closed
Flushing Diesel or Gas	M1 or W1 closed; PXO or XO open
Depressurization	SDV-P, PCK open; M1 or W1 closed; PXO, XO closed

Plausible ranges

Pressure	Temperature	Valve opening	Flow rate	Valve state
0 to 100 MPa	−50 to 250 °C	0 to 100 %	0 to 10 m^3/s	$<10^8$

What each fault does to its signature

Faults & Features pages ► What does a fault do across sensors, and a sensor across faults?

Fault	Inst.	Wells	From	Topside			Seabed		Subsurface
				ABER-CKP	P-MON-CKP	T-JUS-CKP	P-TPT	T-TPT	P-PDG
Normal Operation	594	9							
Abrupt Increase of BSW	4	3	thesis		↓3/3 (1)	↑3/3 (1)	↓4/4	↑4/4	
Spurious Closure of DHSV	22	7	paper		↓6/7 (15)		↓21/21 (1)	↓22/22	↑15/16 (6)
Severe Slugging	32	2	paper		∼	∼	∼27/32		∼4/31 (1)
Flow Instability	343	7	thesis		∼(112)	∼(195)	∼(85)		∼(309)
Rapid Productivity Loss	11	3	thesis		↓11/11	↓10/11	↓6/10 (1)	↓10/11	
Quick Restriction in PCK	6	2	paper	↓0/1 (5)	↑1/3 (3)		↑4/6		↑-(6)
Scaling in PCK	36	6	thesis		↑36/36	↓27/32 (4)	↑29/31 (5)	↓7/32 (4)	
Hydrate in Production Line	14	9	paper		↓9/14		↑7/8 (5)	↓6/7 (7)	↑9/9 (5)
Hydrate in Service Line	57	15	none		?		?(32)	?(27)	?(28)

↑ ↓ the move the 3W paper/thesis expects from the last hour of normal operation to the last hour of the event

∼ a swing

? no source

a/*b* instances that make it, of those that can show it (Severe Slugging: a swing over the paper's 10 bar).

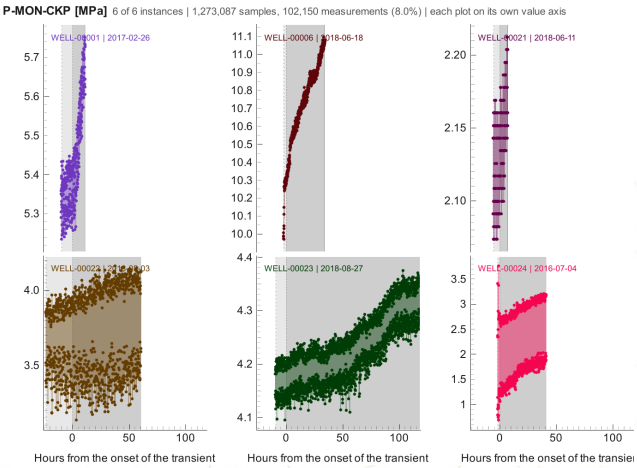
(m) instances where the variable is absent or frozen

Most signatures hold, but each has its exceptions, and some cannot be checked at all

The same fault, well by well

Faults & Features pages ► What does a fault do across sensors, and a sensor across faults?

Scaling in PCK: P-MON-CKP from the onset of the transient, the earliest instance of each well

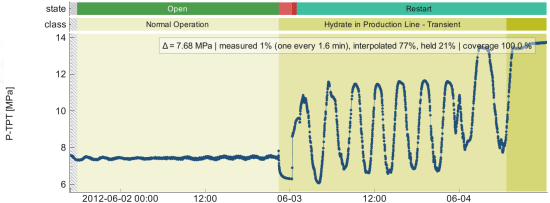


- **Scaling in PCK**: P-MON-CKP rises in 36 of 36, but T-TPT, expected to fall, rises in 15 of WELL-00024's 19 instances.
- **Rapid Productivity Loss**: WELL-00016's P-TPT rises in 4 of 4.
- **Quick Restriction in PCK**: P-PDG is frozen in 6 of 6 instances, the choke opening live in 1.
- **Severe Slugging**: swings P-TPT by 2.6 MPa (median), 27 of 32 over 10 bar, but P-PDG by 0.89 (4 of 31)
- **Flow Instability**: swings P-TPT by 0.07.

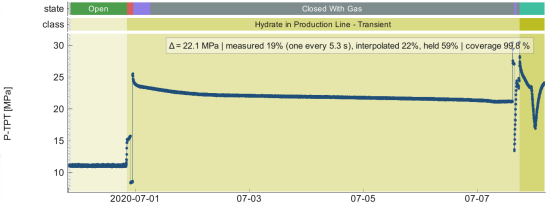
Hydrate in Production Line: the ugly duckling

Faults & Features pages ► What does a fault do across sensors, and a sensor across faults?

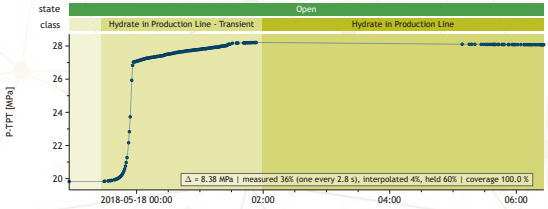
WELL-00019: steady means **plugged**.



WELL-00025: steady means **reopened**.



SIMULATED-00001: steady means **plugged**.



Hydrate labeling across instances, real or simulated, do not seem to share the same philosophy

Appendix

PETROBRAS

Thank you!



github.com/Thiago-TP/3w-well-data-viewer

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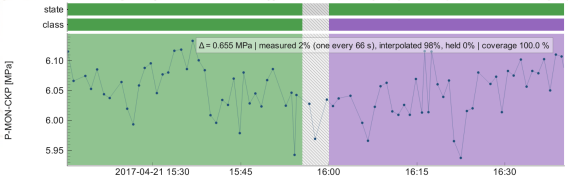
Appendix: label changes across an unlabeled stretch

The 16 unlabeled stretches of a joined history between two different labels.

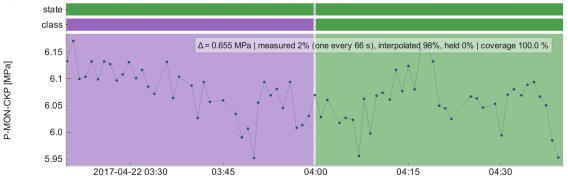
	Well	Unlabeled from	Label before	Label after	For	Shown on
1	WELL-00001	2017-04-21 15:55	normal	Flow Instability (steady)	4 min	P-MON-CKP
2	WELL-00001	2017-04-22 03:59	Flow Instability (steady)	normal	25 s	P-MON-CKP
3	WELL-00001	2017-06-26 09:59	Flow Instability (steady)	normal	3 min	P-MON-CKP
4	WELL-00002	2013-12-12 08:59	normal	Flow Instability (steady)	41 s	P-TPT
5	WELL-00002	2013-12-14 22:00	normal	Flow Instability (steady)	9 s	P-ANULAR
6	WELL-00004	2014-08-06 22:00	Flow Instability (steady)	normal	32 s	P-MON-CKP
7	WELL-00004	2017-10-31 19:14	Quick Restriction in PCK (steady)	normal	5 s	P-MON-CKP
8	WELL-00004	2017-10-31 19:48	Quick Restriction in PCK (steady)	normal	15 s	P-MON-CKP
9	WELL-00006	2018-06-18 11:00	Abrupt Increase of BSW (steady)	normal	30 min	P-MON-CKP
10	WELL-00016	2018-04-26 14:44	Rapid Productivity Loss (transient)	normal	7 min	P-MON-CKP
11	WELL-00020	2014-03-18 01:30	Rapid Productivity Loss (steady)	normal	29 min	P-MON-CKP
12	WELL-00020	2014-03-18 13:57	Rapid Productivity Loss (transient)	normal	53 min	P-MON-CKP
13	WELL-00020	2014-03-19 11:00	Rapid Productivity Loss (transient)	normal	1 h	P-MON-CKP
14	WELL-00024	2016-09-11 07:00	Scaling in PCK (transient)	normal	40 min	P-MON-CKP
15	WELL-00042	2014-12-18 00:53	Hydrate in Service Line (transient)	normal	48 min	P-MON-CKP
16	WELL-00042	2014-12-22 03:37	Hydrate in Service Line (transient)	normal	8 min	P-MON-CKP

Appendix: label changes 1 to 4 (of 16)

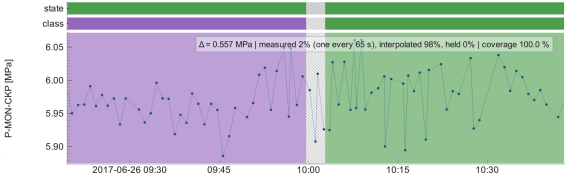
Timelines page ► How are the instances of one well distributed in time?



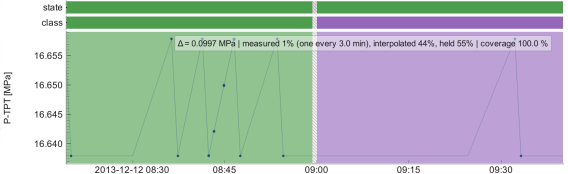
1 WELL-00001, 2017-04-21 15:55: normal → 4 min unlabeled → Flow Instability (steady)



2 WELL-00001, 2017-04-22 03:59: Flow Instability (steady) → 25 s unlabeled → normal



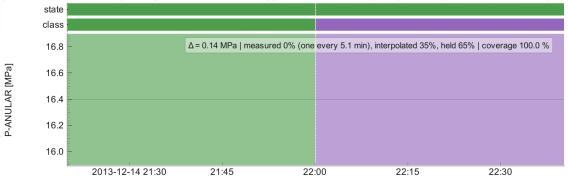
3 WELL-00001, 2017-06-26 09:59: Flow Instability (steady) → 3 min unlabeled → normal



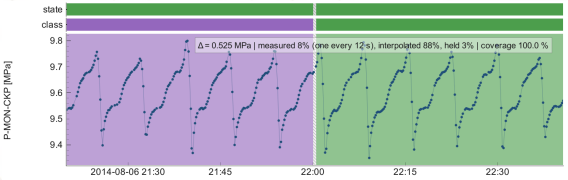
4 WELL-00002, 2013-12-12 08:59: normal → 41 s unlabeled → Flow Instability (steady)

Appendix: label changes 5 to 8 (of 16)

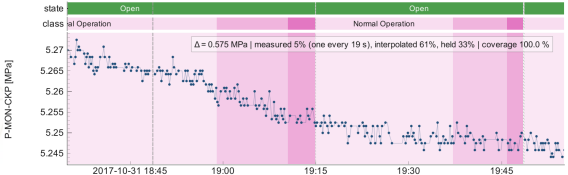
Timelines page ► How are the instances of one well distributed in time?



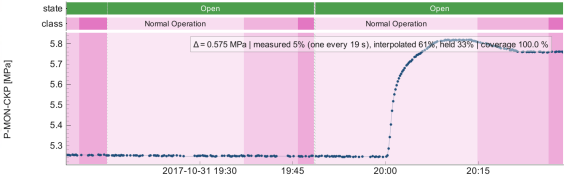
5 WELL-00002, 2013-12-14 22:00: normal → 9 s unlabeled → Flow Instability (steady)



6 WELL-00004, 2014-08-06 22:00: Flow Instability (steady) → 32 s unlabeled → normal



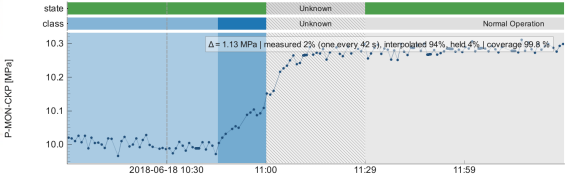
7 WELL-00004, 2017-10-31 19:14: Quick Restriction in PCK (steady) → 5 s unlabeled → normal



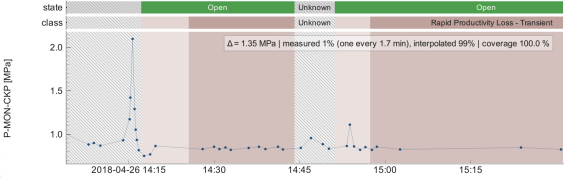
8 WELL-00004, 2017-10-31 19:48: Quick Restriction in PCK (steady) → 15 s unlabeled → normal

Appendix: label changes 9 to 12 (of 16)

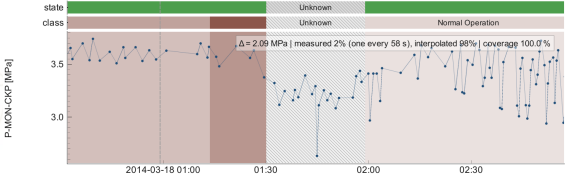
Timelines page ► How are the instances of one well distributed in time?



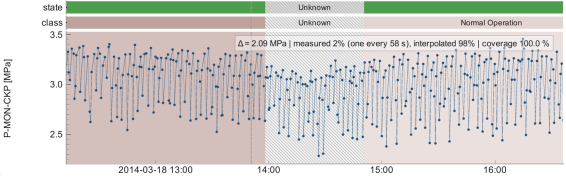
9 WELL-00006, 2018-06-18 11:00: Abrupt Increase of BSW (steady) → 30 min unlabeled → normal



10 WELL-00016, 2018-04-26 14:44: Rapid Productivity Loss (transient) → 7 min unlabeled → normal



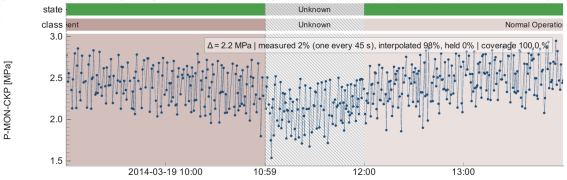
11 WELL-00020, 2014-03-18 01:30: Rapid Productivity Loss (steady) → 29 min unlabeled → normal



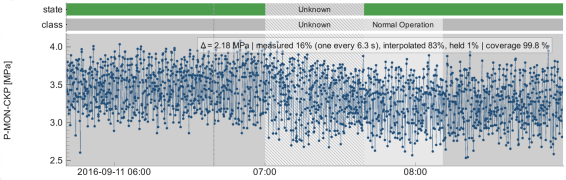
12 WELL-00020, 2014-03-18 13:57: Rapid Productivity Loss (transient) → 53 min unlabeled → normal

Appendix: label changes 13 to 16 (of 16)

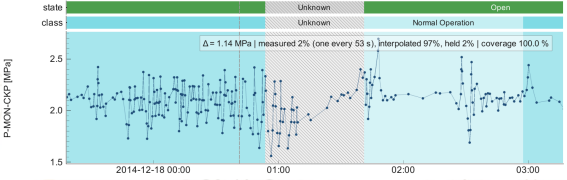
Timelines page ► How are the instances of one well distributed in time?



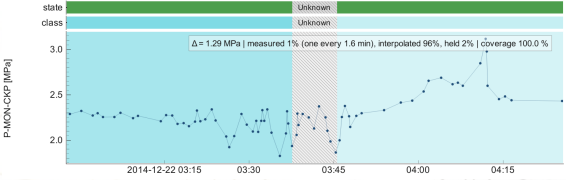
13 WELL-00020, 2014-03-19 11:00: Rapid Productivity Loss (transient) → 1 h unlabeled → normal



14 WELL-00024, 2016-09-11 07:00: Scaling in PCK (transient) → 40 min unlabeled → normal



15 WELL-00042, 2014-12-18 00:53: Hydrate in Service Line (transient) → 48 min unlabeled → normal



16 WELL-00042, 2014-12-22 03:37: Hydrate in Service Line (transient) → 8 min unlabeled → normal

Appendix: what each variable recorded

Availability page ► How well do sensors and states cover the dataset?

Variable	Live			Frozen			Implausible			Out of order			Against state			Missing			Farthest out
	samples	%	inst.	samples	%	inst.	samples	%	inst.	samples	%	inst.	samples	%	inst.	samples	%	inst.	
T-TPT	26,178,605	79.6	1,006	2,236,395	6.8	29	0	0.0	0	-	-	-	-	-	-	4,457,250	13.6	84	
P-TPT	25,878,401	78.7	922	1,612,066	4.9	38	1,004,849	3.1	10	361,705	1.1	29	-	-	-	5,381,783	16.4	159	2942 MPa
P-MON-CKP	25,749,471	78.3	720	42,928	0.1	2	12,489	0.0	5	104,739	0.3	6	-	-	-	7,079,851	21.5	397	−0.237 MPa
P-ANULAR	25,320,751	77.0	914	139,764	0.4	7	387,650	1.2	3	1,296,207	3.9	6	-	-	-	7,411,735	22.5	198	−7.658 MPa
ESTADO-SDV-P	24,316,107	74.0	661	0	0.0	0	0	0.0	0	-	-	-	653,430	2.0	23	8,556,143	26.0	458	
P-JUS-CKGL	24,307,254	73.9	656	743,125	2.3	33	1,299,132	4.0	85	1,298,303	4.0	7	-	-	-	7,821,871	23.8	430	−0.4977 MPa
ESTADO-W1	18,938,036	57.6	558	0	0.0	0	0	0.0	0	-	-	-	3,086,682	9.4	14	13,934,214	42.4	561	
ESTADO-XO	18,538,730	56.4	533	0	0.0	0	0	0.0	0	-	-	-	3,351	0.0	1	14,333,520	43.6	586	
ESTADO-W2	18,293,462	55.7	533	0	0.0	0	0	0.0	0	-	-	-	-	-	-	14,578,788	44.4	586	
ESTADO-PXO	18,176,203	55.3	532	0	0.0	0	0	0.0	0	-	-	-	57,977	0.2	3	14,696,047	44.7	587	
ESTADO-M1	17,773,205	54.1	501	0	0.0	0	0	0.0	0	-	-	-	8,889	0.0	3	15,099,045	45.9	618	
ESTADO-M2	17,688,282	53.8	504	0	0.0	0	0	0.0	0	-	-	-	-	-	-	15,183,968	46.2	615	
ESTADO-SDV-GL	17,105,271	52.0	480	0	0.0	0	0	0.0	0	-	-	-	-	-	-	15,766,979	48.0	639	
T-JUS-CKP	16,689,089	50.8	554	150,261	0.5	8	0	0.0	0	-	-	-	-	-	-	16,032,900	48.8	557	
ESTADO-DHSV	13,865,397	42.2	470	0	0.0	0	0	0.0	0	-	-	-	-	-	-	19,006,853	57.8	649	
QGL	11,150,903	33.9	147	6,569,202	20.0	298	0	0.0	0	-	-	-	-	-	-	15,152,145	46.1	674	
ABER-CKP	9,248,257	28.1	272	3,078,201	9.4	27	161,872	0.5	2	-	-	-	11,955	0.0	4	20,545,792	62.5	820	−99.99 %
P-PDG	7,119,658	21.7	174	17,836,376	54.3	653	1,200,682	3.7	33	256,966	0.8	24	-	-	-	7,916,216	24.1	292	-1.2×10^{36} MPa
T-PDG	6,426,251	19.5	142	14,385,021	43.8	421	2,445,015	7.4	36	-	-	-	-	-	-	12,060,978	36.7	556	-1.7×10^{38} °C
T-MON-CKP	4,789,008	14.6	14	161,538	0.5	1	161,538	0.5	1	-	-	-	-	-	-	27,921,704	84.9	1,104	−99.99 °C
P-JUS-CKP	3,287,363	10.0	10	161,538	0.5	1	1,506,941	4.6	7	0	0.0	0	-	-	-	29,423,349	89.5	1,108	1559 MPa
ABER-CKGL	1,403,126	4.3	48	7,088,387	21.6	238	0	0.0	0	-	-	-	-	-	-	24,380,737	74.2	833	
P-MON-CKGL	308,726	0.9	2	0	0.0	0	6,401	0.0	1	2,096	0.0	1	-	-	-	32,563,524	99.1	1,117	−0.094 MPa
P-JUS-BS	0	0.0	0	0	0.0	0	0	0.0	0	-	-	-	-	-	-	32,872,250	100.0	1,119	
P-MON-SDV-P	0	0.0	0	0	0.0	0	0	0.0	0	-	-	-	-	-	-	32,872,250	100.0	1,119	
PT-P	0	0.0	0	0	0.0	0	0	0.0	0	-	-	-	-	-	-	32,872,250	100.0	1,119	
QBS	0	0.0	0	0	0.0	0	0	0.0	0	-	-	-	-	-	-	32,872,250	100.0	1,119	

Appendix: hydrate instances transients and steadies

Faults & Features pages ► What does a fault do across sensors, and a sensor across faults?

Well	From	Transient			Steady		
		h	state	%	h	state	%
WELL-00019	2012-06-01	32	Restart	94	6	Restart	100
WELL-00019	2014-11-17	73	Shut-In	98	2	Flushing Diesel	100
WELL-00019	2015-06-14	30	Shut-In	92	1	Bullheading	100
WELL-00019	2021-06-17	53	Shut-In	86	2	Flushing Diesel	100
WELL-00025	2020-06-29	165	Closed With Gas	93	11	Restart	100
WELL-00026	2016-06-01	43	Open	100	2	Open	100
WELL-00026	2017-02-07	126	Open	100		never	
WELL-00026	2017-06-08	79	Open	100		never	
WELL-00027	2023-09-18	166	Shut-In	100	8	Flushing Diesel	100
WELL-00028	2021-06-17	76	Shut-In	65	2	Restart	100
WELL-00029	2020-10-20	23	Shut-In	72	1	Bullheading	100
WELL-00030	2014-11-08	33	Shut-In	81	2	Flushing Diesel	100
WELL-00031	2014-04-03	92	Shut-In	90	1	Bullheading	100
WELL-00032	2011-08-30	24	Shut-In	98	2	Flushing Diesel	100
81 simulated			Open	100		Open	100